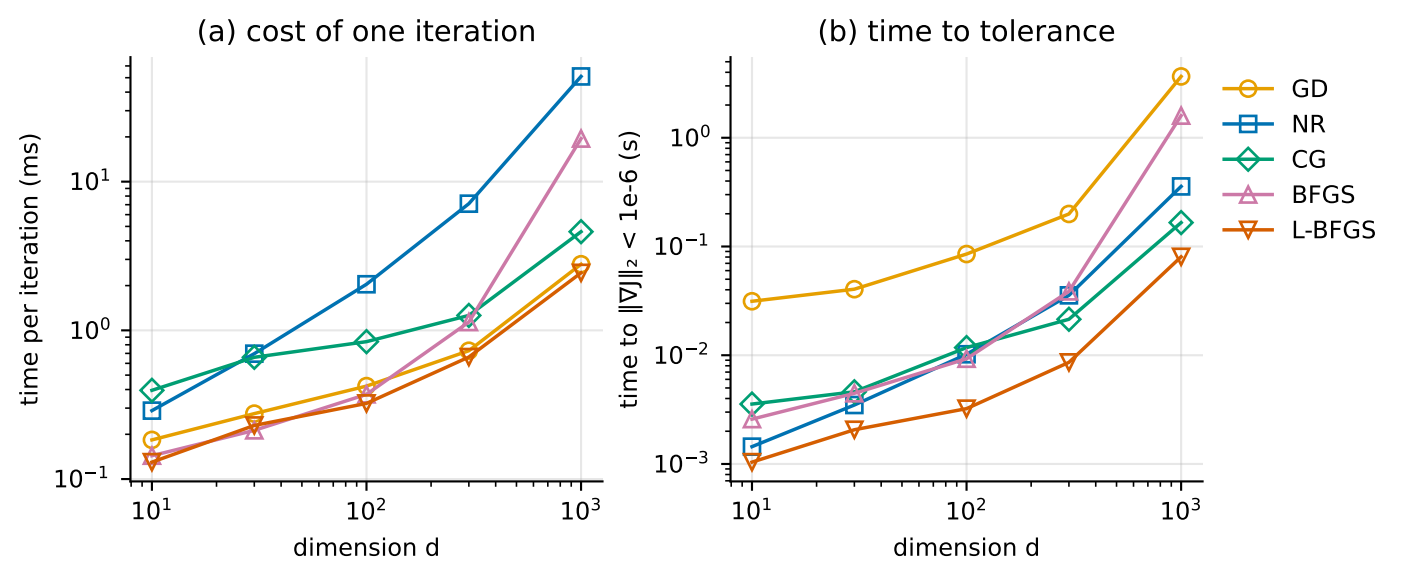
(a) cost of one iteration
time per iteration (ms)
$10^{1}$
$10^{0}$
$10^{-1}$
$10^{1}$
$10^{2}$
$10^{3}$
dimension d
(b) time to tolerance
time to $\|\nabla J\|_2$ < 1e-6 (s)
$10^{0}$
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{1}$
$10^{2}$
$10^{3}$
dimension d
GD
NR
CG
BFGS
L-BFGS